

Network Troubleshooting Lab 002

DNS Failure

Scenario:

A user reports that they are unable to access websites by name (example: google.com), but network connectivity appears to be available. This troubleshooting lab focuses on identifying and resolving DNS-related issues.

Document Information

Category: Networking / IT Support

Lab Type: Troubleshooting Case

Environment: MacBook Pro - macOS

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1. Troubleshooting Methodology

Followed a structured troubleshooting approach:

1. Identify the problem
 2. Establish a theory of probable cause
 3. Test the theory
 4. Analyze results
 5. Document findings
-

2. Step 001 — Verify Network Connectivity

Command Used

ping 8.8.8.8

Purpose

Used to confirm that the device can communicate with an external IP address before troubleshooting DNS.

Evidence

```
Last login: Mon Oct 5 19:32:54 on ttys000
ridleyherisse@Ridleys-MacBook-Pro ~ % ping 8.8.8.8

PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=33.226 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=34.661 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=35.317 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=31.502 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=27.791 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=29.689 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=33.405 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=35.104 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=114 time=42.665 ms
^C
--- 8.8.8.8 ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 27.791/33.700/42.665/3.979 ms
ridleyherisse@Ridleys-MacBook-Pro ~ %
```

Findings

The device successfully communicated with the external IP address, confirming that network connectivity was functioning. This indicated that the issue was likely related to DNS resolution rather than general network connectivity.

The ping test successfully transmitted and received network packets with no packet loss. This confirmed that the device had working network connectivity and allowed troubleshooting to continue toward DNS resolution.

This is actually a good example of the troubleshooting logic:

Ping works + DNS fails = likely DNS issue

Ping fails + DNS fails = broader network connectivity issue

3. Step 002 — Test DNS Resolution

Command Used

nslookup google.com

Purpose

Used to verify whether the DNS server could successfully translate domain names into IP addresses.

Evidence

```
Last login: Mon Oct 5 19:32:54 on ttys000
ridleyherisse@Ridleys-MacBook-Pro ~ % ping 8.8.8.8

PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=33.226 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=34.661 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=35.317 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=31.562 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=27.791 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=29.629 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=33.405 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=35.184 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=114 time=42.665 ms
^C
--- 8.8.8.8 ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 27.791/33.788/42.665/3.979 ms
ridleyherisse@Ridleys-MacBook-Pro ~ % nslookup google.com

Server:          2001:558:feed::1
Address:         2001:558:feed::1#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.211.174

ridleyherisse@Ridleys-MacBook-Pro ~ % █
```

Findings

The DNS lookup successfully resolved the domain name into an IP address, confirming that DNS resolution was functioning correctly.

4. Step 003 — Check DNS Configuration

Command Used

networksetup -getdnsservers Wi-Fi

Purpose

Used to identify the DNS servers configured on the device.

Evidence

```
Last login: Mon Oct 5 19:32:54 on ttys000
ridleylherisse@Ridleys-MacBook-Pro ~ % ping 8.8.8.8

PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=33.226 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=34.661 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=35.317 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=31.582 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=27.791 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=29.629 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=33.485 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=35.184 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=114 time=42.665 ms
^C
--- 8.8.8.8 ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 27.791/33.780/42.665/3.979 ms
ridleylherisse@Ridleys-MacBook-Pro ~ % nslookup google.com

Server:          2001:558:feed::1
Address:         2001:558:feed::1#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.211.174

ridleylherisse@Ridleys-MacBook-Pro ~ % networksetup -getdnsservers Wi-Fi

There aren't any DNS Servers set on Wi-Fi.
ridleylherisse@Ridleys-MacBook-Pro ~ %
```

Findings

The system reported that no manually configured DNS servers were set for the Wi-Fi interface. DNS settings may be provided automatically through DHCP by the network router.

5. Step 004 — Verify Resolution After DNS Correction

Command Used

nslookup google.com

Purpose

Used to confirm successful DNS resolution after troubleshooting.

Evidence

```
Last login: Mon Oct 5 19:32:54 on ttys000
ridleylherisse@Ridleys-MacBook-Pro ~ % ping 8.8.8.8

PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=33.226 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=34.661 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=35.317 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=31.502 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=27.791 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=29.629 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=33.406 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=35.104 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=114 time=42.665 ms
^C
--- 8.8.8.8 ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 27.791/33.700/42.665/3.979 ms
ridleylherisse@Ridleys-MacBook-Pro ~ % nslookup google.com

Server:      2001:558:feed::1
Address:     2001:558:feed::1#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.211.174

ridleylherisse@Ridleys-MacBook-Pro ~ % networksetup -getdnsservers Wi-Fi
There aren't any DNS Servers set on Wi-Fi.
ridleylherisse@Ridleys-MacBook-Pro ~ % nslookup google.com

Server:      2001:558:feed::1
Address:     2001:558:feed::1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.68.206

ridleylherisse@Ridleys-MacBook-Pro ~ % █
```

Findings

DNS resolution was restored, confirming that the device could successfully translate domain names into IP addresses.

6. Conclusion

The troubleshooting process identified DNS resolution as the source of the issue. DNS configuration was reviewed and verified to restore proper domain name resolution.

7. Skills Demonstrated

- DNS Troubleshooting
- TCP/IP Fundamentals
- Network Diagnostics
- Command Line Tools
- Root Cause Analysis
- IT Documentation